

ZONE 4: REVIEW - API ENDPOINTS SPECIFICATION

3. API ENDPOINTS

Base URL: `/api/v1/review`

3.1 AFTER-ACTION REPORTS

Generate After-Action Report

typescript

TypeScript

POST /api/v1/review/reports/generate

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_id: string;
  event_id: string;
  report_type: 'automatic' | 'comprehensive' | 'executive' |
'compliance';
  include_recommendations?: boolean;
  include_benchmarks?: boolean;
  include_evidence_ledger?: boolean;
  distribution_list?: string[]; // User IDs or email addresses
  schedule_distribution?: {
    send_at?: Date;
    channels?: ('email' | 'dashboard' | 'api' | 'pdf')[];
  };
}
```

Response: 201 Created

```
{
  success: true;
  data: {
    report_id: string;
    status: 'generating';
    estimated_completion_seconds: number;
    tracking_url: string;
  };
  metadata: {
    request_id: string;
    timestamp: string;
  };
}
```

Response: 400 Bad Request

```
{
  success: false;
  error: {
    code: 'INVALID_EVENT';
    message: 'Event must be completed before generating report';
    details: {
      event_status: string;
      event_end_time: string;
    };
  };
}
```

Get Report Status

typescript

TypeScript

```
GET /api/v1/review/reports/{report_id}/status
Authorization: Bearer {token}
```

```
Response: 200 OK
{
  success: true;
  data: {
    report_id: string;
    status: 'generating' | 'review' | 'approved' | 'published';
    progress_percentage: number; // 0-100
    current_stage: string; // 'data_collection', 'analysis',
    'report_generation', etc.
    estimated_completion: string; // ISO timestamp
    errors?: string[];
  };
}
```

Get After-Action Report

typescript

```
TypeScript
GET /api/v1/review/reports/{report_id}
Authorization: Bearer {token}
Query Parameters:
  - format?: 'json' | 'pdf' | 'html'
  - include_evidence?: boolean
  - include_raw_data?: boolean

Response: 200 OK (format=json)
{
  success: true;
  data: AfterActionReport; // Full report object
  metadata: {
    generated_at: string;
    report_version: string;
    data_sources: string[];
  };
}
```

```
}
```

```
Response: 200 OK (format=pdf)
Content-Type: application/pdf
Content-Disposition: attachment;
filename="AAR-{venue}-{event}-{date}.pdf"
[Binary PDF data]
```

List Reports

typescript

TypeScript

```
GET /api/v1/review/reports
Authorization: Bearer {token}
Query Parameters:
  - venue_id?: string
  - event_id?: string
  - report_type?: string
  - status?: string
  - start_date?: string (ISO date)
  - end_date?: string (ISO date)
  - page?: number (default: 1)
  - limit?: number (default: 20, max: 100)
  - sort_by?: 'event_date' | 'created_at' | 'report_status'
  - sort_order?: 'asc' | 'desc'
```

```
Response: 200 OK
{
  success: true;
  data: {
    reports: AfterActionReport[];
    pagination: {
      current_page: number;
      total_pages: number;
      total_count: number;
    }
  }
}
```

```

        has_next: boolean;
        has_prev: boolean;
    };
};
}

```

Approve Report

typescript

TypeScript

```

POST /api/v1/review/reports/{report_id}/approve
Authorization: Bearer {token}
Content-Type: application/json

```

Request Body:

```

{
  approval_notes?: string;
  publish_immediately?: boolean;
  distribution_override?: string[];
}

```

Response: 200 OK

```

{
  success: true;
  data: {
    report_id: string;
    status: 'approved';
    approved_by: string;
    approved_at: string;
    published: boolean;
  };
}

```

Publish Report

typescript

TypeScript

```
POST /api/v1/review/reports/{report_id}/publish
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  distribution_channels: ('email' | 'dashboard' | 'api' |
'pdf')[];
  recipients: string[];
  notification_message?: string;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    report_id: string;
    published_at: string;
    distribution_summary: {
      total_recipients: number;
      channels_used: string[];
      delivery_status: {
        delivered: number;
        pending: number;
        failed: number;
      };
    };
  };
};
}
```

3.2 PERFORMANCE METRICS & ANALYTICS

Get Event Performance Metrics

typescript

TypeScript

GET /api/v1/review/performance/events/{event_id}

Authorization: Bearer {token}

Query Parameters:

- include_benchmarks?: boolean
- include_historical?: boolean
- compare_to_event_id?: string

Response: 200 OK

```
{
  success: true;
  data: {
    event_metrics: EventPerformanceMetrics;
    benchmarks?: {
      league_comparison: ComparisonData[];
      historical_comparison: ComparisonData[];
      peer_comparison: ComparisonData[];
    };
    insights: {
      key_achievements: string[];
      improvement_areas: string[];
      notable_changes: string[];
    };
  };
}
```

Analyze Performance Trends

typescript

TypeScript

POST /api/v1/review/performance/trends

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
```

```
venue_id: string;
start_date: string;
end_date: string;
metrics: string[]; // Metric names to analyze
aggregation?: 'daily' | 'weekly' | 'monthly';
include_predictions?: boolean;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    trends: {
      [metric_name: string]: {
        dates: string[];
        values: number[];
        trend_direction: 'improving' | 'stable' | 'declining';
        change_percentage: number;
        statistical_significance: number;
        predictions?: {
          next_period_value: number;
          confidence_interval: [number, number];
          forecast_accuracy: number;
        };
      };
    };
    summary: {
      improving_metrics: string[];
      declining_metrics: string[];
      stable_metrics: string[];
      overall_trend: string;
    };
  };
}
```

Compare Venue Performance

typescript

TypeScript

POST /api/v1/review/performance/compare

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_ids: string[];
  period_start: string;
  period_end: string;
  metrics: string[];
  normalize?: boolean; // Normalize for venue size/capacity
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    comparison: {
      [metric_name: string]: {
        venues: {
          venue_id: string;
          venue_name: string;
          value: number;
          rank: number;
        }[];
        league_average: number;
        best_practice_value: number;
      };
    };
    insights: {
      top_performers: { venue_id: string; metric: string; value:
number }[];
      improvement_opportunities: { venue_id: string; metric:
string; gap: number }[];
    };
  };
}
```

```
};  
}
```

Get Performance Dashboard

typescript

TypeScript

```
GET /api/v1/review/performance/dashboard/{venue_id}  
Authorization: Bearer {token}  
Query Parameters:  
  - period?: 'current_month' | 'last_30_days' | 'current_quarter'  
  | 'custom'  
  - start_date?: string (required if period=custom)  
  - end_date?: string (required if period=custom)
```

Response: 200 OK

```
{  
  success: true;  
  data: {  
    summary: {  
      overall_score: number; // 0-100  
      total_events: number;  
      total_incidents: number;  
      incidents_prevented: number;  
      avg_response_time_seconds: number;  
      guest_satisfaction_avg: number;  
      compliance_score: number;  
    };  
    category_scores: {  
      safety: number;  
      operations: number;  
      efficiency: number;  
      satisfaction: number;  
    };  
    trends: {
```

```
[metric_name: string]: {
  current_value: number;
  previous_value: number;
  change_pct: number;
  sparkline_data: number[]; // Last 10 data points
};
};
alerts: {
  declining_metrics: string[];
  improvement_opportunities: string[];
  notable_achievements: string[];
};
};
}
```

3.3 INCIDENT ANALYSIS

Analyze Incident

typescript

TypeScript

POST /api/v1/review/incidents/{incident_id}/analyze

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  analysis_type: 'root_cause' | 'response_effectiveness' |
'prevention' | 'comprehensive';
  include_similar_incidents?: boolean;
  generate_recommendations?: boolean;
}
```

Response: 201 Created

```
{
  success: true;
  data: {
    analysis_id: string;
    status: 'analyzing';
    estimated_completion_seconds: number;
  };
}
```

Get Incident Analysis

typescript

```
TypeScript
GET /api/v1/review/incidents/{incident_id}/analysis
Authorization: Bearer {token}
Query Parameters:
  - analysis_id?: string (get specific analysis)
  - include_evidence?: boolean

Response: 200 OK
{
  success: true;
  data: IncidentAnalysis;
}
```

Bulk Analyze Incidents

typescript

```
TypeScript
POST /api/v1/review/incidents/analyze-bulk
Authorization: Bearer {token}
Content-Type: application/json

Request Body:
{
```

```
incident_ids?: string[]; // Specific incidents
event_id?: string; // All incidents from event
venue_id?: string; // Requires date range
start_date?: string;
end_date?: string;
filters?: {
  severity?: string[];
  category?: string[];
  preventability?: string[];
};
}
```

Response: 202 Accepted

```
{
  success: true;
  data: {
    job_id: string;
    total_incidents: number;
    estimated_completion_minutes: number;
    status_url: string;
  };
}
```

Get Pattern Analysis

typescript

TypeScript

GET /api/v1/review/patterns

Authorization: Bearer {token}

Query Parameters:

- venue_id: string (required)
- pattern_type?: string
- confidence_threshold?: number (0-100, default: 70)
- start_date?: string
- end_date?: string

```
- limit?: number
```

Response: 200 OK

```
{
  success: true;
  data: {
    patterns: PatternAnalysis[];
    summary: {
      total_patterns: number;
      critical_patterns: number;
      high_confidence_patterns: number;
      recommended_actions: string[];
    };
  };
}
```

Detect New Patterns

typescript

TypeScript

POST /api/v1/review/patterns/detect

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_id: string;
  start_date: string;
  end_date: string;
  pattern_types?: string[];
  confidence_threshold?: number;
  ml_enabled?: boolean;
}
```

Response: 202 Accepted

```

{
  success: true;
  data: {
    job_id: string;
    analysis_scope: {
      events_analyzed: number;
      incidents_analyzed: number;
      date_range: { start: string; end: string };
    };
    estimated_completion_minutes: number;
  };
}

```

3.4 IMPROVEMENT RECOMMENDATIONS

Create Recommendation

typescript

TypeScript

POST /api/v1/review/recommendations

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```

{
  venue_id: string;
  recommendation_title: string;
  recommendation_description: string;
  recommendation_type: 'process' | 'technology' | 'training' |
'policy' | 'infrastructure';
  priority_level: 'critical' | 'high' | 'medium' | 'low';
  urgency: 'immediate' | 'short_term' | 'medium_term' |
'long_term';
  estimated_impact: 'high' | 'medium' | 'low';
}

```

```
    source_incident_id?: string;
    source_analysis_id?: string;
    estimated_cost?: number;
    estimated_hours?: number;
    required_resources?: string[];
    success_criteria?: string[];
  }
```

Response: 201 Created

```
{
  success: true;
  data: ImprovementRecommendation;
}
```

List Recommendations

typescript

TypeScript

GET /api/v1/review/recommendations

Authorization: Bearer {token}

Query Parameters:

- venue_id?: string
- status?: string
- priority?: string
- type?: string
- assigned_to?: string
- page?: number
- limit?: number
- sort_by?: 'priority' | 'created_at' | 'target_date'
- sort_order?: 'asc' | 'desc'

Response: 200 OK

```
{
  success: true;
  data: {
```

```

    recommendations: ImprovementRecommendation[];
    summary: {
      total: number;
      by_status: { [status: string]: number };
      by_priority: { [priority: string]: number };
      overdue: number;
    };
    pagination: PaginationMetadata;
  };
}

```

Update Recommendation Status

typescript

TypeScript

PATCH /api/v1/review/recommendations/{recommendation_id}/status

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```

{
  status: 'proposed' | 'approved' | 'in_progress' | 'completed' |
'rejected' | 'deferred';
  notes?: string;
  implementation_owner?: string;
  target_date?: string;
}

```

Response: 200 OK

```

{
  success: true;
  data: ImprovementRecommendation;
}

```

Approve Recommendation

typescript

TypeScript

POST /api/v1/review/recommendations/{recommendation_id}/approve

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  approval_notes?: string;
  assigned_to?: string;
  budget_allocated?: number;
  target_completion_date?: string;
  priority_override?: 'critical' | 'high' | 'medium' | 'low';
}
```

Response: 200 OK

```
{
  success: true;
  data: ImprovementRecommendation;
}
```

Get Recommendation ROI Analysis

typescript

TypeScript

GET /api/v1/review/recommendations/{recommendation_id}/roi

Authorization: Bearer {token}

Response: 200 OK

```
{
  success: true;
  data: {
    recommendation_id: string;
    estimated_costs: {
      implementation: number;
      training: number;
      ongoing_maintenance: number;
    };
  };
}
```

```

        total: number;
    };
    estimated_benefits: {
        cost_savings_annual: number;
        efficiency_gains_hours: number;
        risk_reduction_value: number;
        revenue_impact: number;
        total_annual: number;
    };
    roi_analysis: {
        payback_period_months: number;
        roi_percentage: number;
        five_year_npv: number;
        break_even_date: string;
    };
    confidence_level: 'high' | 'medium' | 'low';
    assumptions: string[];
};
}

```

3.5 IMPLEMENTATION TRACKING

Create Implementation

typescript

```

TypeScript
POST /api/v1/review/implementations
Authorization: Bearer {token}
Content-Type: application/json

```

Request Body:

```

{
  recommendation_id: string;
  implementation_plan: string;
}

```

```
    implementation_approach: 'phased' | 'immediate' | 'pilot' |
    'full_rollout';
    project_lead: string;
    team_members: string[];
    budget_allocated: number;
    hours_estimated: number;
    milestones: Milestone[];
    rollback_plan?: string;
  }
```

Response: 201 Created

```
{
  success: true;
  data: ImprovementImplementation;
}
```

Update Implementation Progress

typescript

TypeScript

```
PATCH /api/v1/review/implementations/{implementation_id}/progress
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
    progress_status: 'planning' | 'executing' | 'testing' |
    'validating' | 'completed';
    percent_complete: number; // 0-100
    current_milestone?: string;
    budget_spent?: number;
    hours_spent?: number;
    progress_notes?: string;
    blockers?: string[];
  }
```

```
Response: 200 OK
{
  success: true;
  data: ImprovementImplementation;
}
```

Complete Milestone

typescript

```
TypeScript
POST
/api/v1/review/implementations/{implementation_id}/milestones/{milestone_id}/complete
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  completion_notes?: string;
  actual_completion_date?: string;
  evidence_refs?: string[];
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    milestone: Milestone;
    implementation_progress: number; // Overall % complete
    next_milestone?: Milestone;
  };
}
```

Run Validation Tests

typescript

TypeScript

```
POST /api/v1/review/implementations/{implementation_id}/validate
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  validation_tests: {
    test_name: string;
    test_description: string;
    test_procedures: string[];
    success_criteria: string[];
  }[];
  schedule_date?: string; // If not immediate
}
```

Response: 202 Accepted

```
{
  success: true;
  data: {
    validation_id: string;
    scheduled_date: string;
    total_tests: number;
    estimated_duration_minutes: number;
  };
}
```

Get Implementation Dashboard

typescript

TypeScript

```
GET /api/v1/review/implementations/dashboard/{venue_id}
Authorization: Bearer {token}
Query Parameters:
  - status?: string
  - priority?: string
```

```
Response: 200 OK
{
  success: true;
  data: {
    summary: {
      total_implementations: number;
      in_progress: number;
      completed: number;
      overdue: number;
      total_budget_allocated: number;
      total_budget_spent: number;
      total_roi_realized: number;
    };
    active_implementations: ImprovementImplementation[];
    upcoming_milestones: Milestone[];
    overdue_items: {
      implementations: ImprovementImplementation[];
      milestones: Milestone[];
    };
    recent_completions: ImprovementImplementation[];
  };
}
```

3.6 BENCHMARKING & COMPARISON

Get Benchmarks

typescript

```
TypeScript
GET /api/v1/review/benchmarks
Authorization: Bearer {token}
Query Parameters:
  - venue_id: string (required)
```

```
- benchmark_type: 'league' | 'division' | 'conference' |  
'venue_tier' | 'custom'  
- metric_category?: string  
- metric_name?: string  
- period_start: string (ISO date)  
- period_end: string (ISO date)
```

Response: 200 OK

```
{  
  success: true;  
  data: {  
    benchmarks: BenchmarkMetric[];  
    venue_summary: {  
      overall_percentile: number;  
      top_performing_metrics: string[];  
      improvement_needed_metrics: string[];  
      competitive_position: 'leader' | 'above_average' |  
'average' | 'below_average';  
    };  
  };  
}
```

Generate Benchmark Report

typescript

TypeScript

POST /api/v1/review/benchmarks/report

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{  
  venue_id: string;  
  benchmark_type: 'league' | 'division' | 'peer_group';  
  period_start: string;
```

```
period_end: string;
metrics: string[]; // Specific metrics or 'all'
include_trends?: boolean;
include_recommendations?: boolean;
format?: 'json' | 'pdf';
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    report_id: string;
    venue_performance: {
      overall_ranking: number;
      total_venues: number;
      percentile: number;
      category_rankings: {
        [category: string]: {
          rank: number;
          score: number;
          percentile: number;
        };
      };
    };
    comparisons: BenchmarkComparison[];
    trends: {
      [metric: string]: {
        current_period: number;
        previous_period: number;
        change_pct: number;
        relative_to_league: 'improving_faster' |
'improving_slower' | 'declining';
      };
    };
    recommendations?: string[];
  };
};
```

```
}
```

Compare to Peer Venues

typescript

TypeScript

```
POST /api/v1/review/benchmarks/peer-comparison
```

```
Authorization: Bearer {token}
```

```
Content-Type: application/json
```

Request Body:

```
{
  venue_id: string;
  peer_venue_ids?: string[]; // If not provided, system selects
similar venues
  metrics: string[];
  period_start: string;
  period_end: string;
  normalization?: 'capacity' | 'revenue' | 'none';
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    venue_performance: { [metric: string]: number };
    peer_averages: { [metric: string]: number };
    detailed_comparison: {
      [metric: string]: {
        venue_value: number;
        venue_rank: number;
        peer_values: {
          venue_id: string;
          venue_name: string;
          value: number;

```

```

        }[];
        best_practice_venue: string;
        best_practice_value: number;
    };
};
insights: {
    strengths: string[];
    opportunities: string[];
    learning_opportunities: {
        metric: string;
        best_practice_venue: string;
        potential_improvement: number;
    }[];
};
};
}

```

3.7 BEST PRACTICES & KNOWLEDGE SHARING

Get Best Practices

typescript

TypeScript

GET /api/v1/review/best-practices

Authorization: Bearer {token}

Query Parameters:

- category?: string
- validation_status?: string
- implementation_difficulty?: string
- min_adoption_count?: number
- sort_by?: 'adoption_count' | 'roi' | 'validation_status'

Response: 200 OK

```
{
```

```
success: true;
data: {
  best_practices: BestPractice[];
  summary: {
    total_practices: number;
    by_category: { [category: string]: number };
    by_validation: { [status: string]: number };
    most_adopted: BestPractice[];
    highest_roi: BestPractice[];
  };
};
}
```

Submit Best Practice

typescript

TypeScript

POST /api/v1/review/best-practices

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  practice_title: string;
  practice_description: string;
  practice_category: string;
  originated_from_venue_id?: string;
  originated_from_event_id?: string;
  implementation_difficulty: 'easy' | 'moderate' | 'complex';
  estimated_implementation_time: string;
  required_resources: string[];
  implementation_steps: any;
  impact_metrics?: {
    cost_savings?: number;
    efficiency_gain_pct?: number;
  };
}
```

```
satisfaction_improvement?: number;
safety_improvement?: string;
};
documentation_links?: string[];
}
```

Response: 201 Created

```
{
  success: true;
  data: BestPractice;
}
```

Adopt Best Practice

typescript

TypeScript

POST /api/v1/review/best-practices/{practice_id}/adopt

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_id: string;
  implementation_plan?: string;
  target_completion_date?: string;
  assigned_to?: string;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    adoption_id: string;
    practice: BestPractice;
    implementation_created: boolean;
  }
}
```

```
    implementation_id?: string;
  };
}
```

Report Best Practice Results

typescript

TypeScript

POST /api/v1/review/best-practices/{practice_id}/results

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_id: string;
  success: boolean;
  actual_cost_savings?: number;
  actual_efficiency_gain?: number;
  actual_satisfaction_improvement?: number;
  lessons_learned: string[];
  implementation_notes: string;
  would_recommend: boolean;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    practice_id: string;
    validation_updated: boolean;
    new_validation_success_rate: number;
  };
}
```

3.8 EVIDENCE LEDGER INTEGRATION

Query Evidence Ledger

typescript

TypeScript

POST /api/v1/review/evidence/query

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  query_type: 'incident_timeline' | 'decision_audit' |
'compliance_evidence' | 'performance_data';
  venue_id: string;
  event_id?: string;
  incident_id?: string;
  start_time?: string;
  end_time?: string;
  filters?: {
    action_types?: string[];
    user_ids?: string[];
    systems?: string[];
  };
  include_metadata?: boolean;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    query_id: string;
    ledger_transactions: {
      transaction_id: string;
      timestamp: string;
      action_type: string;
      actor: string;
      system: string;
    }
  }
}
```

```

        data_hash: string;
        metadata?: any;
    }[];
    summary: {
        total_records: number;
        time_span: { start: string; end: string };
        data_integrity_verified: boolean;
    };
};
}

```

Verify Evidence Chain

typescript

TypeScript

POST /api/v1/review/evidence/verify

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```

{
  transaction_ids: string[];
  verification_type: 'integrity' | 'completeness' |
'chain_of_custody';
}

```

Response: 200 OK

```

{
  success: true;
  data: {
    verification_result: 'verified' | 'failed' | 'incomplete';
    verified_transactions: number;
    failed_transactions: number;
    integrity_score: number; // 0-100
    issues?: {

```

```
        transaction_id: string;
        issue_type: string;
        description: string;
    }[];
};
}
```

Export Evidence Package

typescript

TypeScript

POST /api/v1/review/evidence/export

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  venue_id: string;
  event_id?: string;
  incident_id?: string;
  report_id?: string;
  start_time?: string;
  end_time?: string;
  format: 'json' | 'pdf' | 'blockchain_export';
  include_media?: boolean;
  certification_required?: boolean;
}
```

Response: 202 Accepted

```
{
  success: true;
  data: {
    export_id: string;
    estimated_completion_minutes: number;
    download_url_available_at: string; // URL to check status
  }
}
```

```
};  
}
```

3.9 CONTINUOUS IMPROVEMENT WORKFLOWS

Create Workflow

typescript

TypeScript

```
POST /api/v1/review/workflows  
Authorization: Bearer {token}  
Content-Type: application/json
```

Request Body:

```
{  
  venue_id: string;  
  workflow_name: string;  
  workflow_type: 'incident_review' | 'quarterly_review' |  
'compliance_audit' | 'performance_review';  
  trigger_type?: 'event_based' | 'scheduled' | 'manual' |  
'threshold';  
  trigger_condition?: string;  
  recurrence_pattern?: string;  
  workflow_owner: string;  
  participants: string[];  
  approvers: string[];  
  stages: WorkflowStage[];  
  due_date?: string;  
}
```

Response: 201 Created

```
{  
  success: true;  
  data: ImprovementWorkflow;
```

```
}
```

Start Workflow

typescript

TypeScript

```
POST /api/v1/review/workflows/{workflow_id}/start
```

```
Authorization: Bearer {token}
```

```
Content-Type: application/json
```

Request Body:

```
{
  context?: {
    event_id?: string;
    report_id?: string;
    incident_ids?: string[];
  };
  priority_override?: 'high' | 'normal' | 'low';
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    workflow_id: string;
    started_at: string;
    current_stage: string;
    next_actions: string[];
    assigned_participants: string[];
  };
}
```

Update Workflow Stage

typescript

TypeScript

```
PATCH /api/v1/review/workflows/{workflow_id}/stages/{stage_id}
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  status: 'pending' | 'in_progress' | 'completed' | 'skipped';
  notes?: string;
  completed_by?: string;
  artifacts?: string[]; // URLs or IDs of generated documents
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    workflow: ImprovementWorkflow;
    stage_updated: WorkflowStage;
    next_stage?: WorkflowStage;
    workflow_completed: boolean;
  };
}
```

Complete Workflow

typescript

TypeScript

```
POST /api/v1/review/workflows/{workflow_id}/complete
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  outcome_summary: string;
  action_items: string[];
}
```

```

    follow_up_required?: boolean;
    follow_up_date?: string;
    artifacts?: string[];
  }

Response: 200 OK
{
  success: true;
  data: {
    workflow_id: string;
    completed_at: string;
    total_duration_hours: number;
    action_items_created: number;
    recommendations_generated: number;
    next_scheduled_occurrence?: string; // If recurring
  };
}

```

Get Active Workflows

typescript

```

TypeScript
GET /api/v1/review/workflows/active
Authorization: Bearer {token}
Query Parameters:
  - venue_id?: string
  - workflow_type?: string
  - assigned_to?: string
  - overdue_only?: boolean

Response: 200 OK
{
  success: true;
  data: {
    active_workflows: ImprovementWorkflow[];
  };
}

```

```

summary: {
  total_active: number;
  overdue: number;
  due_this_week: number;
  by_type: { [type: string]: number };
};
};
}

```

3.10 CONTINUOUS IMPROVEMENT DASHBOARD

Get Improvement Dashboard

typescript

TypeScript

```

GET /api/v1/review/dashboard/{venue_id}
Authorization: Bearer {token}
Query Parameters:
  - period?: 'current_month' | 'current_quarter' | 'current_year'
  | 'custom'
  - start_date?: string
  - end_date?: string

```

Response: 200 OK

```

{
  success: true;
  data: ContinuousImprovementDashboard; // Full dashboard object
}

```

Get Improvement Insights

typescript

TypeScript

```

POST /api/v1/review/insights/generate

```

```
Authorization: Bearer {token}
Content-Type: application/json
```

Request Body:

```
{
  venue_id: string;
  insight_types: ('performance_trends' | 'pattern_detection' |
'benchmark_comparison' | 'recommendation_priorities')[];
  period_start: string;
  period_end: string;
  ai_powered?: boolean;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    insights: {
      type: string;
      priority: 'critical' | 'high' | 'medium' | 'low';
      title: string;
      description: string;
      supporting_data: any;
      recommended_actions: string[];
      estimated_impact: {
        cost_savings?: number;
        efficiency_gain?: number;
        risk_reduction?: string;
      };
    }[];
    action_plan: {
      immediate_actions: string[];
      short_term_actions: string[];
      long_term_initiatives: string[];
    };
  };
};
```

```
}
```

3.11 INTEGRATION ENDPOINTS

Sync with PREPARE Zone

typescript

TypeScript

```
POST /api/v1/review/sync/prepare
```

```
Authorization: Bearer {token}
```

```
Content-Type: application/json
```

Request Body:

```
{
  venue_id: string;
  recommendations: string[]; // Recommendation IDs to sync back to
  blueprints
  best_practices: string[]; // Best practice IDs to integrate
  update_blueprints?: boolean;
  update_compliance_frameworks?: boolean;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    blueprints_updated: number;
    compliance_frameworks_updated: number;
    playbooks_enhanced: number;
    sync_summary: string;
  };
}
```

Export to External Systems

typescript

TypeScript

POST /api/v1/review/export/external

Authorization: Bearer {token}

Content-Type: application/json

Request Body:

```
{
  export_type: 'after_action_report' | 'performance_metrics' |
'improvement_recommendations';
  target_system: 'email' | 'slack' | 'teams' | 'jira' |
'servicenow' | 'custom_webhook';
  config: {
    [key: string]: any; // System-specific configuration
  };
  report_id?: string;
  venue_id?: string;
  event_id?: string;
}
```

Response: 200 OK

```
{
  success: true;
  data: {
    export_id: string;
    target_system: string;
    delivery_status: 'sent' | 'queued' | 'failed';
    external_reference?: string; // External system's ID
  };
}
```